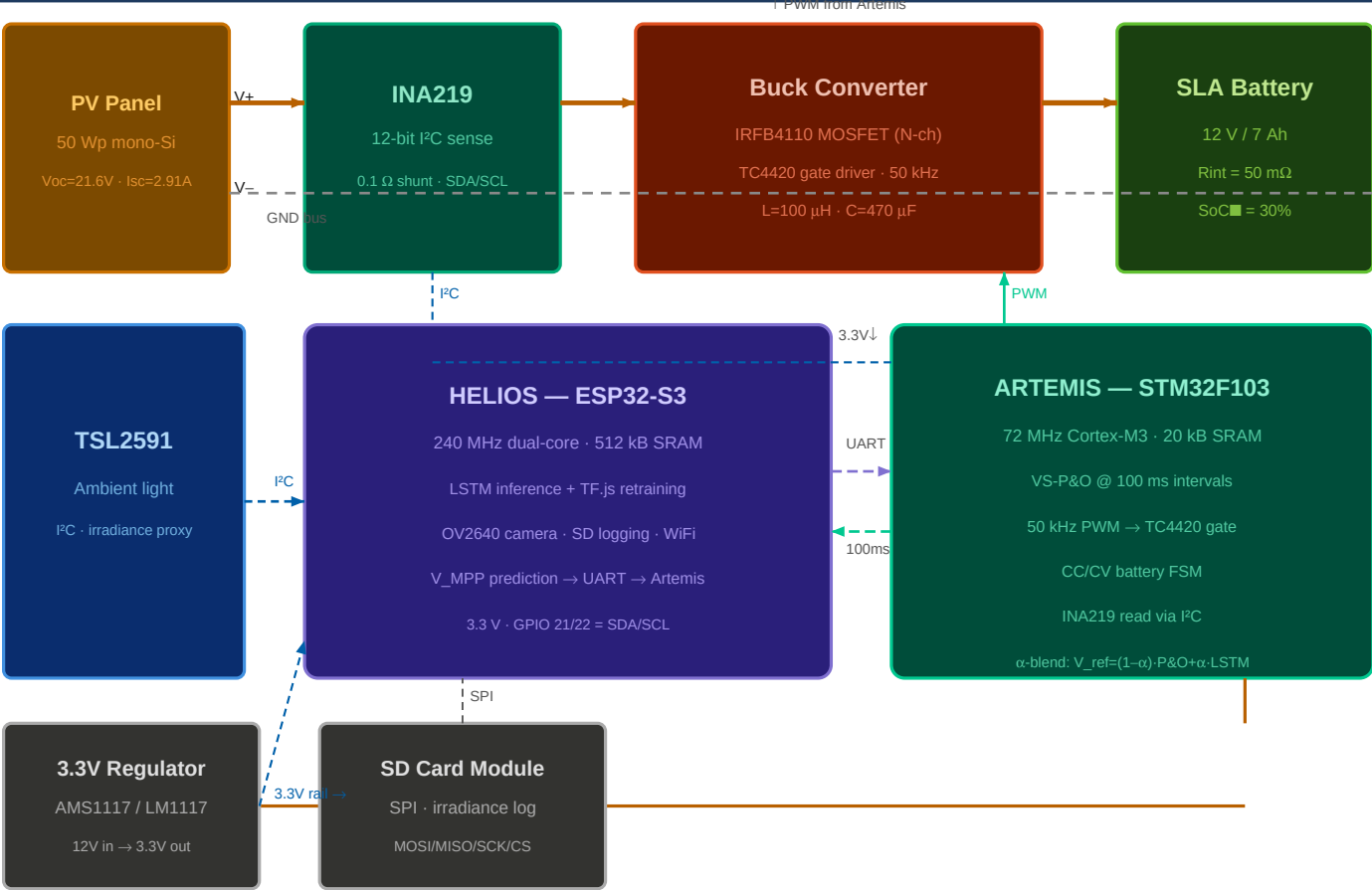
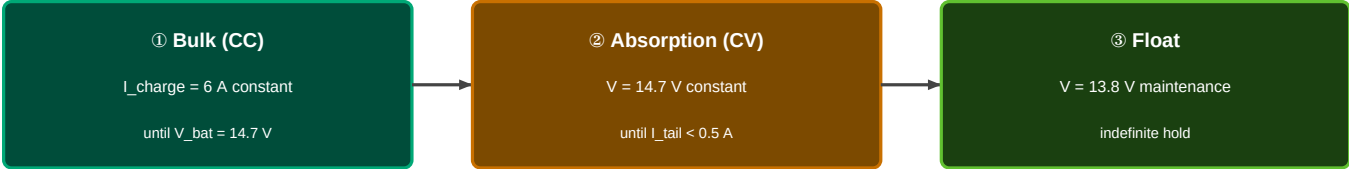




CC/CV Three-Stage Charging — Artemis FSM



LSTM – P&O Blend Law (Artemis control)

$$V_{ref} = (1 - \alpha) \cdot V_{ref,P\&O} + \alpha \cdot V_{MPP,pred}$$

$\alpha \in [0.20, 0.55]$ · optimal $\alpha \approx 0.35$ · fires when █ – G/G > 0.15 · 20-step cooldown

Wiring Reference — Key Connections

Signal	From pin	To pin	Note
I ² C SDA	ESP32-S3 GPIO 21	INA219 SDA · TSL2591 SDA	4.7 kΩ pull-up
I ² C SCL	ESP32-S3 GPIO 22	INA219 SCL · TSL2591 SCL	4.7 kΩ pull-up
STM32 I ² C	STM32 PB6/PB7	INA219 shared bus	addr 0x40
UART TX/RX	ESP32 U1TX (GPIO17)	STM32 PA10 (USART1 RX)	115200 baud
PWM out	STM32 PA8 (TIM1_CH1)	TC4420 IN → IRFB4110 gate	50kHz · 10kΩ series R
SPI SD	ESP32 VSPI (GPIO 23/19/18/5)	SD MOSI/MISO/SCK/CS	SPI mode 0

HELIOS-ARTEMIS — Component Bill of Materials

Ref	Part / Value	Description	Notes	BDT
U1	ESP32-S3	MCU — Helios (AI inference)	240 MHz · 512 kB SRAM · WiFi	380
U2	STM32F103C8T6	MCU — Artemis (real-time ctrl)	72 MHz Cortex-M3 · 20 kB SRAM	120
U3	INA219	Current/voltage sense	12-bit I²C · 0.1 Ω shunt	80
U4	TC4420	MOSFET gate driver	6A peak · 50 kHz rated	—
U5	AMS1117-3.3	3.3V LDO regulator	1A · SOT-223	30
U6	TSL2591	Ambient light / irradiance sensor	I²C · 600 Mlux dynamic range	120
Q1	IRFB4110	N-channel power MOSFET	VDS=100V · RDS(on)=3.7 mΩ · TO-220	—
D1	SS34 Schottky	Freewheeling diode	3A · 40V · DO-214AB	—
L1	100 μH inductor	Buck filter inductor	I_sat ≥ 8A · DCR < 50 mΩ	—
C1	470 μF / 35V	Buck output filter cap	Electrolytic · low-ESR	—
C2	1000 μF / 25V	Input bulk capacitor	Electrolytic · near Q1 drain	—
R1	0.1 Ω / 2W	INA219 shunt resistor	4-wire Kelvin connection	—
R2,R3	4.7 kΩ	I²C pull-up resistors	SDA + SCL to 3.3V	—
R4	10 Ω	Gate series resistor	Limits dV/dt · near TC4420 output	—
J1	2-pin terminal	PV panel input	5A rated · screw terminal	—
J2	2-pin terminal	Battery output	10A rated · screw terminal	—
J3	SD card module	Data logging (SPI)	MOSI/MISO/SCK/CS	—
—	PCB + passives	PCB fabrication + misc R/C/housing	FR4 140×100mm · 2-layer	630

TOTAL CONTROLLER BOM≈ 1,360 BDT (89% below commercial MPPT)

Assembly & Ordering Notes:

- IRFB4110 and TC4420 may be combined cost: source locally or substitute with IRF3205 + TC4427.
- All SMD passives 0603; electrolytic caps must be 105°C rated for outdoor Sylhet conditions.
- INA219 breakout modules include shunt resistor — confirm 0.1 Ω shunt on board before ordering R1.
- ESP32-S3 DevKit-C includes USB-UART — use for programming. STM32 programmed via SWD (ST-Link).
- Order PCB from JLCPCB/PCBWay: 140×100mm, 2-layer, FR4 1.6mm, HASL, green solder mask.

• Place 100nF bypass cap at each IC VCC pin, close to pin — critical for EMC at 50kHz switching.